Theoretical Astrophysics I: Stellar Structure and Evolution
Lecture 7: Simple models of stellar structure

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Recap

- Now, we understand the main processes that govern stellar structure *(not yet evolution!)*.
- To recap our main assumptions:
- Stars are in hydrostatic equilibrium.
- Energy is transported either by radiation or convection
- Hopefully we understand equation of state (connection between T, ρ, P)
- Hopefully we can calculate the opacity from there
- And hopefully we can calculate the rate of energy production.

Equations of stellar structure

Here they are, written in terms of r as independent variable:

$$\frac{dP}{dr} = -\rho\frac{Gm}{r^2} \tag{1}$$

$$\frac{dm}{dr} = 4\pi r^2\rho \tag{2}$$

$$\frac{dT}{dr} = -\frac{3}{4ac}\frac{\kappa\rho}{T^3}\frac{F}{4\pi r^2} \tag{3}$$

$$\frac{dF}{dr} = 4\pi r^2\rho q \tag{4}$$

$$P = \frac{\mathcal{R}}{\mu_I}\rho T + P_e + \frac{1}{3}aT^4 \tag{5}$$

$$\kappa = \kappa_0\rho^a T^b \tag{6}$$

$$q = q_0\rho^m T^n \tag{7}$$

Equations of stellar structure

Count the unknowns, count the equations, is the system, in principle, solvable?

$$
\begin{aligned}
\frac{dP}{dr} &= -\rho\frac{Gm}{r^2} \\
\frac{dm}{dr} &= 4\pi r^2\rho \\
\frac{dT}{dr} &= -\frac{3}{4ac}\frac{\kappa\rho}{T^3}\frac{F}{4\pi r^2} \\
\frac{dF}{dr} &= 4\pi r^2\rho q \\
P &= \frac{\mathcal{R}}{\mu_I}\rho T + P_e + \frac{1}{3}aT^4 \\
\kappa &= \kappa_0\rho^a T^b \\
q &= q_0\rho^m T^n
\end{aligned}
$$

Equations of stellar structure

Four of these equations are differential equations. Three are algebraic. Differential equations are of the first order, so each one needs one boundary condition. These are, generally speaking.

- At $r = 0$ (center), $m = 0$, $F = 0$.
- At $m = M$, $P = 0$ and $F = L$.

It turns out that two physical parameters that govern the structure a so-called ZAMS (zero age main sequence star) star are the **total mass** and **chemical composition**. More about that in the next class.

Can someone tell me why we could not say at $r = R$, $m = M$ and $P = 0$ and be done with it?

Equations of stellar structure

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Can someone tell me why we could not say at $r = R$, $m = M$ and $P = 0$ and be done with it? Because we do not know what R is? When solving the equations of stellar structure, we don't know in advance how big the star will be!!!

Some simple assumptions that can be made

To ease our job, some simple things can, generally be assumed

- Constant μ, either for ZAMS or completely evolved stars.
- When $q =$, F is constant.
- P and ρ and T can all be safely assumed to monotoneously decrease with r (m).
- It is super helpful if we can find combinations of quantities that stay approximately constant over the stellar radius.

Now - be mindful that the situation can be even more dire - as the opacity and nuclear rate production can be very complicated and no analytical solution is possible.
In modeling, in general, we solve everything by numerics.